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Factors Affecting Strategic Inventory Control And Operational Performance Of Manufacturing Firms In Daynile District Mogadishu, Somalia

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Abstract

The effectiveness of strategic inventory control strategies and operational performance may be measured in the demanding and competitive world of today. The purpose of this study was to examine the factors affecting strategic inventory control and operational performance of manufacturing firms in Daynile district Mogadishu Somalia. Descriptive design was the method used, with the target population of manufacturing entities in Daynile district. Information from primary (questionnaires) and secondary sources was combined during this study. The study participants were 31 respondents thus census was carried out to study the entire population of 34 employees of manufacturing firms in Daynile district. According to the findings, EOQ, Just-In-Time, ABC Analysis, Warehouse Management System, and Demand Forecasting were all found to have been greatly adopted by the Manufacturing Companies in the Daynile district of Somalia. Descriptive statistics were used to analyze the first objective and regression analysis was used to analyze the second objective. The results also show that the dependent variable (operational performance) was positively influenced by the independent variable (strategic inventory control) and that the adoption of strategic inventory control significantly impacted efficiency, speed, and quality. Manufacturing companies are advised to heavily employ strategic inventory control techniques. Further studies are also recommended in the future.

Introduction

Research is constructed on theories as its foundation. They include a group of associated ideas that the researcher will use to determine what will be measured in order to establish a relationship between variables. The theoretical framework establishes the topic's value by giving the reader the knowledge they need to comprehend the study. Academic academics and other practitioners have not placed a strong emphasis on performance measurement procedures. Yet, organizational performance has always had an impact on how organizations operate and make decisions. "You can't manage it if you can't measure it" (Garvin 1993). The pertinent theories that explain inventory management control techniques and perform the best were presented in this section. The Transaction Cost Analysis (TCA) Theory and the Stochastic Inventory Theory were the theories being investigated.

Transactional Cost Analysis Theory (TCA)

A theory starting in economics is transaction cost analysis (TCA), often known as transaction cost theory or transaction cost economics. TCA seeks to describe policy choices. The official and informal rules of exchange are referred to as governance in the context of transaction costs. Early TCA research mostly addressed the make-or-buy decision of enterprises and covered issues including whether firms should use their own sales force or engage with independent agents, as well as whether components should be sourced or made internally. The definition of governance has been broadened by recent TCA work to cover how buy choices are made using contracts or social mechanisms like trust, partner selection, or enforcement. Moreover, efforts to pick partners, outsourcing, the use of crowdsourcing, and other similar activities are governance, and may therefore be explained using TCA (Ghosh and John, 1999).

The mathematical economist Williamson developed the transactional cost analysis theory (TCA) in 1974, which came into being in the 1970s. When one party finds it challenging to evaluate the performance of the other before, during, or after the transaction, measurement costs result. For instance, maintaining the standard of outsourced advertising, legal counsel, or Technology involves significant fees. TCAT has grown to be a key anchor for the examination of a variety of organizational and strategic issues that are very crucial to a corporation (Williamson, 2006). In strategic inventory management, Wigand (2005) opined that TCT is crucial in determining the various costs, such as stockholding costs, ordering costs, carrying costs, and price per unit of inventory that make up a product's constituent costs and enable optimal resource allocation and serve as a safety net against the organization incurring unnecessary costs, which in turn improves performance. TCAT is beneficial because it provides the final total cost of a product after production, allowing manufacturers to plan their spending based on the order fulfillment technique they will adopt, allowing them to meet the demand for increasingly diverse products (Pine, 2006). Similar to this, TCAT enables manufacturing companies to maintain different classes of inventories based on their dollar value to a company that, in turn, balances the needs of the customer with the capabilities of the supply chain by primarily forecasting on demand and synchronizing it with production, procurement, and distribution capabilities to enable timely delivery of supplies in the right quantity (Gonzalez, 2010).

Stochastic Inventory Theory

In order to efficiently meet the demand for commodities, stochastic inventory theory addresses the management of stock levels. Determine the time of issuing replenishment orders and the related order quantity subject to demand uncertainty and/or other system characteristics is the goal of stochastic

inventory control Porteus (2002). By Evan (2002) is about the control of stock levels with the explicit goal of supplying customers with the products kept in inventory. By using mathematical models to describe the inventory system that determines inventory policy, businesses are able to save money. According to Laan, Kiesmüller, Kuik, Vlachos, and Dekker (2004), physical transfer of the items from the supplier to the customer is necessary to satisfy the demand for a certain volume of commodities. Lack of goods results from unfulfilled orders for patient clients, backlogged demand, or lost sales as a result of impatient customers (Laan et al., 2004).

The stochastic models, according to Beamon et al. (2006), are more important since they reflect reality better. By taking into account the costs of stock shortfalls, stock arrangement, and stock storage, they attempt to create the best inventory plan possible. In the current study, the theory is relevant as it will aid the manufacturing firms in ways that they can be able to avoid shortages of inventory and be able to meet customer demand on time. The theory will also help the firm in knowing how to keep records of stock when to replenish the stock and how much to replenish to meet the demands of their clients effectively and efficiently (Laan et al., 2004).

Strategic Inventory Control

According to Worldwide, As the lifeblood and the core of every production system, inventory is viewed as being essential to the smooth operation of any manufacturing company (Ballou, 2005). Inventory frequently accounts for up to 30% of all capital invested by industrial enterprises (Dobler, 2006). Sawaya (2006) made this observation and suggested that it might be responsible for up to 90% of the working capital and 33% of the company's assets. According to Pandey (2005), Companies must implement a strategic inventory control process to determine the amount of merchandise that must be kept in their warehouses at all times. The company's policies, practices, processes, and controls are to be included in the control mechanisms in order to monitor the level of inventory that should be stocked. As they decide the quantity of inventory to be stocked and the manipulation of the current stock in the stores, the implemented control mechanisms aid in the appropriate management of inventory.

An inventory issue results from the management of possessions in entities. Strategic inventory control techniques are mostly used to minimize costs or maximize profits in order to maintain happy customers. Excess inventory takes up space, increases the risk of damage, causes economic problems, and increases the likelihood of loss and spoiling. The use of strategic inventory control techniques helps to ensure that the aforementioned issues are avoided by the businesses. The following strategic inventory control techniques were discussed in this paper: Just-In-Time (JIT), ABC Analysis, Warehouse Management System (WMS), Economic Order Quantity (EOQ), and Demand Forecasting (Kagira 2012)

Operational performance

Operational Performance (OP) is the process of evaluating an organization's performance in relation to predetermined or established measures of environmental responsibility, efficacy, and effectiveness (Vencataya, 2011). Operational performance concentrates on a company's internal working capability with regard to cost reduction and waste minimization, product quality enhancement and generating new products, enhanced delivery capacity as well as increased productivity as explained by (Riyadi & Munizu, 2013). The efficacy and efficiency of firm's operations are frequently assessed against set benchmarks for productivity, waste reduction, and regulatory compliance (Slack et al., 2004). Organizations should pay attention to operational performance since it improves production efficiency, produces high-quality goods, creates delighted customers, and leads to increased profits and revenues for the entity (Kaynak & Harley, 2008). The company's

operational performance defines how well it produces goods and services efficiently and effectively, as well as how well those goods and services meet customers' expectations and needs (Vencataya, 2011). Hwang, Han, Jun and Park (2014) view The management of an entity's processes depends on operational performance, which is also essential in laying the foundation for sustainable competitiveness. It also helps the business function better so that it can achieve its fundamental goals, such as service delivery, quality, and productivity (Bayo-Moriones & De Cerio, 2002).

Hwang et al. (2014) point out that the Assessment of operational performance is a fundamental task and a crucial part of maintaining a firm's competitive advantage. Birech (2011) further posits that utilization, lead time, measuring inventory, cycle time, measuring quality, time, speed, cost, efficiency, and effectiveness are a few of the methods used to determine operational performance. According to Vencataya, Keshwar, and Deveshika (2015), there are various performance optimization goals, such as efficiency, quality, and speed. Thus, these operational performance measures were used in the study. The emphasis on operational performance has been focused on Efficiency, Quality, and Speed given the broad scope of this study. Businesses must think about how to use all of their resources effectively and efficiently if they want to achieve high operational performance. Employers must view workers as assets that must be used in the most effective and efficient manner in order to maximize production and employee motivation. Efficiency refers to making the optimum use of a resource set in order to maximize gains (Vencataya, 2011).

Manufacturing firms of Daynile district

Before the civil war, Somalia had a thriving industrial history, with fifty-three small, medium, and big businesses managing the industry. The nation had the chance to create juices, textiles, canned meat, fruit, and fish, food items, sugar, alcoholic drinks, some cosmetics, beverages, and candies, among other things. In contrast to the past and the other countries in the region, it is today extremely low and few. According to data from Somalia's customs, imports of capital equipment for manufacturing and raw materials have increased during the last three years. This shows that Somalia's manufacturing sector has a bright future because to the nation's entrepreneurial culture. Due to an unstable and harsh climate and a cap on private investments, manufacturing is still slow. Due to these dangers, it was challenging to draw in investors for current technology (World Bank, 2018).

Methodology

The dataset used in this study has been obtained from manufacturing firms in the Daynile district. The study was descriptive and cross-sectional in design. it is cross-sectional in nature because it intends to collect data at a given point in time from a cross-section of respondents of manufacturing firms in Daynile district. The study was adopted a descriptive research design because it intends to describe factors affecting strategic inventory control and operational performance of manufacturing firms in Daynile district, Mogadishu – Somalia. When gathering data, the researcher also utilized a quantitative technique. The study population consisted of 34 respondents from manufacturing companies headquartered in the Daynile district of Mogadishu that have been registered by the government's Ministry of Commerce and Industry. Respondents from manufacturing enterprises in the Daynile area were the study's target population. The researcher worked with a sample size of 31 respondents (91.2%) selected from the target population respondents of manufacturing firms in Daynile district.

This research sampling procedure was purposive sampling which is a form of non-probability sampling. In this study, the questionnaire was the main instrument of this research. The study adopted structured questionnaire as the main tool for the collection of primary data from the target

population administered by trained data collectors and the researcher and all interviewers were provided with appropriate written materials about factors affecting strategic inventory control and operational performance of manufacturing firms in Daynile district to introduce the objective and to obtain information from respondents to complete the questionnaires without any problems. The study respondents were the supply chain officers, inventory personnel, warehouse and equivalent positions. The questionnaire was structured into three sections namely: Section A which constituted the background information, section B that covered Strategic inventory control Techniques adopted by manufacturing companies in Daynile district and section C which constituted metrics of operating performance that are linked to the adoption of Strategic inventory control techniques of manufacturing firms in Daynile district Mogadishu Somalia.

$Y = a + \beta 1X1 + \beta 2X2 + \beta 3X3 + \beta 4X4 + \beta 5X5 + e$

Were,

Y=Operational Performance

X_1=Economic order quantity

X_2= Just-In-Time

X_3= ABC analysis

X_4= Warehouse Management System

X_5= Demand Forecasting

a = constant

β1 β2 β3 β4 β5 are the coefficients of the independent variables X_1, X_2, X_3, and X_4, respectively and e is the error term.

Results

After receiving the completed surveys, the gathered data was reviewed for biases, inaccuracies in the responses, omissions, and exaggerations. Quantitative Primary data from the questionnaire was analyzed concurrently to avoid duplication thereby guiding the entire study for balanced and critical analysis using the statistical package for social sciences (SPSS) version 20. Data was interred into SPSS and the first objective was analyzed by the use of descriptive statistics through the use of mean, mode, median and standard deviation. The second objective of the study was analyzed by the use of regression analysis to bring out the relationship that exists between the variables. The subsequent regression analysis equation that was used is;

Introduction

This chapter presents the data analysis results, interpretations and discussion of Strategic Inventory Control Techniques adoption and the regression analysis showing the relationship between Strategic Inventory Control Techniques and Operational Performance.

Response rate

Out of a total number of 34 respondents that received the self-administered questionnaires, 31 respondents answered and returned to the researcher. This gave a positive response rate of 91.2% and a non-response rate of 8.8%. This was a good representative sample of the targeted population of manufacturing firms in Daynile district and helped the researcher to

arrive at the right conclusions.

Demographic information

This was divided based on the respondent's positions in the organization, length of service in their current position and the period which the Manufacturing firms have been operational in Mogadishu, Somalia.

Table: 4.2.1. Age of respondents

Age of respondents	Frequency	Percent
less than 20 years	5	16.1
21-30 years	15	48.4
31-40 years	9	29.0
41-50 years	2	6.5
Total	31	100.0

Table4.2.1: shows the age grouping of the respondents the majority of the respondents 15 (48.4%) were aged between 21-30 years, the next age group 9 (29.0%) were aged between 31-40 years, 5(16.1%) of the respondents were aged less than 20 years finally the least of the respondents 2(6.5%) were aged 41-50 years. This indicates the major numbers who responded questions were between the age of 21 -30 years and are active manpower that work in different departments of manufacturing firms in Daynile district.

Table 1.2.2: Gender

Gender of respondents	Frequency	Percent
Male	4	12.9
Female	27	87.1
Total	31	100.0

Table 1.2.2: indicates gender of the respondents, the majority of respondents 27 (87.1%) were males while the rest of respondents 4 (12.9%) were females, so the result outcome showed the highest respondents were males who are the most manufacturers of manufacturing companies in Daynile district than females who are the least proportion of these respondents but work in managerial sections of manufacturing firms.

Table 1.2.3: Marital status

Marital status of respondents	Frequency	Percent
Single	8	25.8
Married	14	45.2
Divorced	6	19.4
Widow	3	9.6
Total	31	100.0

Table 1.2.3: indicates marital status of the respondents, the majority of the respondents 14 (45.2%) were married, 8 (25.8%) were single, 6(19.4%) were divorced, while the remaining least of respondents 3 (9.6) were widowed. therefore, this implies that the majority of the respondents were married and have the desire of manufacturing to cover their household needs and supply the district population's needs.

Table 1.2.5: Level of education

Educational level	Frequency	Percent
primary level	2	6.5
secondary level	8	25.8
college or university	21	67.7
Total	92	100.0

Table 1.2.5: indicates the level of education of the respondents, the majority of the respondent 21(67.7%) were university level, the next showed that 8 (25.8%) were secondary level, finally 2 (6.5%) were primary level. This implies that the majorities of the respondents were university level and have knowledge of manufacturing and strategic control of their inventories and operation performance.

General Information

Table 1.2.1: Positional category in manufacturing firms

Positional category in the manufacturing firms	Frequency	Percent
Supply chain managers	8	25.80
Warehouse Managers	12	38.70
Procurement Officers	4	12.90
Logistics Managers	7	22.60
Total	31	100.0

Table 1.2.1: shows that 25.80% of the respondents were supply chain managers, 38.70% were Warehouse Managers, 12.90% were Procurement Officers while the other 22.60% of respondents were Logistics Managers. The findings conclude that majority of the respondents (87.1%) were in managerial positions (Supply Chain Managers, Warehouse Managers and Logistics Managers) and were better suited to give information that can be relied upon and had knowledge on the subject under study.

Table 1.2.2: length of service in years

Length of service(years)	Frequency	Percent
1-2	5	16.13
3-5	8	25.81
5-10	11	35.48
Over 10	7	22.58
Total	31	100.0

Table 1.2.2: indicates the length of service, 16.13% of the respondents had worked in the manufacturing firms for less than two years (1-2). 25.81% had worked for periods between 3-5 years while 35.48% had worked for 5-10 years. The remaining 22.58% had worked for 10 years and above. Thus 83.87% of the respondents had worked for more than three years which shows that they had experience and knowledge of answering the questionnaires.

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Period of Existence

The respondents were asked to answer for how long their Manufacturing firms had been operational and the Table 1.3.1 below shows their responses

Table 1.3.1 Manufacturing firm Existence

Period of existence (years)	Frequency	Percent
1-5	10	32.26
5-10	12	38.71
Over 10	9	29.03
Total	31	100.0

Table 1.3.1: The findings indicated that 32.26% of the Manufacturing firms have been operational for a period of one to five years while 38.71 % have been operating for a period between five to ten years and the remaining 29.03% have been operational for over ten years. This indicates that majority of the Manufacturing firms (67.74%) have been operational for more than five years which is a significant duration showing that they were qualified to give relevant feedback to this study. This indicates that they have been long enough in the field to understand the best Strategic Inventory Control Techniques.

4.4 Extent of Strategic Inventory Control Adoption

The study sought to determine the extent to extent.

which the Manufacturing firms in Mogadishu had adopted Strategic Inventory Control Techniques. The rating was done on a Likert scale where 1= to a very small extent, 2= to a small extent, 3= to a medium extent, 4= to a large extent and 5= to a very large extent. The subsequent sub sections present the Strategic Inventory Control Techniques which were further subdivided into five categories of Economic Order Quantity, Just- in-Time, ABC Analysis, Warehouse Management System and Demand Forecasting. The tables below summarize the mean scores with a low margin above 3 and average of 4/5 to a very large extent affirming that Manufacturing firms have adopted Strategic Inventory Control Techniques from medium to a large

Economic Order Quantity	**Mean**	**Std. Dev**
The organization adopts a classical economic order quantity model	3.52	1.06
The organization observes periodical replenishment of stocks	3.71	0.86
The organization maintains minimum stock levels	3.55	1.12
The organization adopts EOQ Model to minimize storage	4.00	0.89
Overall score	**3.69**	**1.08**

Table 1.4.1: Economic Order Quantity

Table 1.4.1: indicates respondents to a large extent (M= 4.00, SD= 0.89) agreed that the Manufacturing firms adopts EOQ Model to minimize storage**.** Periodical stock replenishment (M= 3.71, SD= 0.86), maintaining Minimum Stock Levels (M= 3.55, SD= 1.12) and adopting a classical EOQ Model

(M= 3.52, SD= 1.06) were adopted to a large extent as reflected by their respective mean and standard deviation. Overall score indicates that Economic Order Quantity has been adopted to a large extent by the manufacturing firms as shown by the mean of 3.69 and standard deviation of 1.08.

Table 1.4.2: Just-in-Time

Just-in-Time	Mean	Std. Dev
The organization uses JIT stock control system	3.87	0.88
The organization uses the JIT system to eliminate waste	3.52	1.06
The organization uses JIT to maintain enough materials in just the right place and time	3.71	0.86
The organization uses JIT to reduce mishandling of raw materials	3.55	1.12
Overall score	**3.66**	**0.75**

Table 1.4.2: use of Stock Control System (Mean= 3.87, SD= 0.88), maintaining enough materials in the right place and at the right time (Mean= 4.05, SD= 1.07) and having a proper information flow within the organization (Mean= 3.71, SD= 0.86) were all adopted to a large extent. Use of JIT to reduce mishandling of raw materials with a mean of 3.55 and Standard Deviation of 1.12 and using JIT to eliminate waste (Mean= 3.52, SD=1.06), were also adopted to a large extent. The overall score (Mean= 3.66, SD= 0.75) shows that Just-In-Time was adopted to a large extent by the Manufacturing firms in Daaynile district Mogadishu-Somalia.

Table 1.4.3: Activity Based Computing (ABC) Analysis

ABC Analysis	Mean	Std. Dev
The organization uses ABC analysis to classify items according to their stock value	4.00	0.89
The organization uses ABC Analysis to reduce stock holding cost	3.97	0.88
The organization uses ABC Analysis to manage stock levels	4.02	0.97
The organization uses ABC Analysis to aid in setting of the prices strategically for different classification of items	4.01	0.91
Overall score	**4.00**	**0.88**

Table 1.4.3: shows that the use of ABC to manage stock levels (Mean= 4.02, SD= 0.97), use of ABC in setting prices for classified items (Mean= 4.01, SD= 0.91) and classifying items based on their stock value (Mean= 4.00, SD= 0.89) were all adopted to a large extent. The

use of ABC to reduce stock holding cost was adopted to a large extent as shown by mean of 3.97 and standard deviation of 0.88.

Overall score reflects that ABC Analysis has been adopted by manufacturing firms to a large extent as shown by the mean of 4.00 and SD of 0.88. These findings show that ABC analysis is a well-established categorization technique based on the Pareto Principle for determining which items should get priority in the management of a company's inventory and implementation of ABC analysis is key in the management of inventory in manufacturing firms and facilitates cutting of cost through proper attention given specific classes of inventory in the order of priorit

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Table 1.4.4: Warehouse Management System

Warehouse Management System	Mean	Std. Dev
Warehouse Management System helps in reviewing inventory levels on a timely basis	4.29	0.87
Warehouse Management System aids in detecting stock variance	3.91	1.06
Warehouse Management System helps in managing inbound and outbound logistics	4.13	0.86
Warehouse Management System helps in utilizing store space Efficiently	4.25	0.81
Overall score	**4.15**	**0.87**

Table 1.4.4: presents Warehouse Management System techniques that have been adopted by manufacturing firms in Mogadishu and the response indicates that WMS aids in reviewing inventory levels on a timely basis to a large extent (Mean=4.29, SD= 0.87). WMS in managing outbound and inbound logistics (Mean=4.13, SD= 0.88) and WMS use in utilizing store space efficiently (Mean=4.25, SD= 0.81) have also been adopted to a large extent. Use of WMS in detecting stock variance has been adopted to a large extent as indicated by the mean of 3.91 and standard deviation of 1.06. Warehouse Management System has been adopted to a large extent (Mean=4.15, SD= 0.87) as reflected by the overall score.

Table 1.4.5: Demand Forecasting

Demand Forecasting	Mean	Std. Dev
We ensure that we keep records for future reference	3.54	1.12
We anticipate the demand for our products in future	3.91	0.89
We estimate sales during a specified future period based on the proposed marketing plan	3.87	0.88
We take into account forecast errors while forecasting demand	3.55	1.12
Overall score	**3.72**	**1.33**

Table 1.4.5: Shows that Demand Forecasting Techniques where anticipating demand for future products was adopted to a large extent with a mean of 3.91 and SD of 0.89 while estimating sales for specified future period based on proposed marketing plan was also adopted to a large extent with the mean of 3.87 and SD of 0.88. keeping records for future reference (Mean=3.54, SD= 1.12) and taking into account forecast error (Mean=3.55, SD= 1.12) has been also been adopted to a large extent. The overall score (Mean=3.72, SD= 1.33) means that Demand Forecasting Techniques have been adopted to a large extent by Manufacturing firms in Daynile district Mogadishu.

4.5 Operational Performance Outcomes of Adopting Strategic Inventory Control

The respondents had to give feedback on how Strategic Inventory Control Techniques affect the Operational Performance of their respective Manufacturing Firms. The outcomes focused on Efficiency, Quality and Speed. The Tables below summarizes the mean scores with a low margin above 3 and an average of 4/5 who agreed that Strategic Inventory Control influences Operational Performance

Table 1.5.1: Strategic Inventory Control and Efficiency

Efficiency	**Mean**	**Std. Dev**
The firm has improved inventory turns	4.47	0.97
The firm has improved throughput	4.19	1.09
The firm has improved schedule/ production attainment	3.94	1.01
The firm has improved capacity utilization	3.83	1.18
The firm has improved overall equipment effectiveness	4.11	0.93
Overall Score	**4.10**	**0.89**

Table 1.5.1: shows that improved inventory turnover (M= 4.47, SD= 0.97) , improved through put (M= 4.19, SD= 1.09) and improved equipment effectiveness (M= 4.11, SD= 0.93) were influenced by Strategic Inventory Control to a large extent as shown by their respective means and SD. Improved production attainment (M= 3.94, SD= 1.012) and improved capacity utilization (M= 3.83, SD= 1.18) were influenced to a large extent as well. The overall score indicates that Strategic Inventory Control Techniques Influence the Efficiency of Manufacturing Firms in Daynile district Mogadishu to a large extent as shown by the mean of 4.10 and SD of 0.89.

Table 1.5.2: Strategic Inventory Control and Speed

Speed	Mean	Std. Dev
Reduced cycle time	4.21	0.85
Predictable throughput time	4.13	1.06
Increased delivery frequency	4.05	1.10
Reduced time in serving customers	3.97	0.98
Increased order handling time	4.09	0.99
Overall Score	**4.08**	**0.92**

Table 1.5.2: shows that Strategic Inventory Control Techniques influence a firms speed to a large extent through reduced cycle time (M= 4.21, SD= 0.85), predictable through put time (M= 4.13, SD= 1.06) and increased delivery frequency (M= 4.05, SD= 1.10) as indicated by the respective mean and SD. Increased order handling time (M= 4.09, SD= 0.99) and reduced time in serving customers (M= 3.97, SD= 0.98) also influence performance to a large extent. The overall score (M= 4.08, SD= 0.92) indicates that Strategic Inventory Control Techniques influences Speed of Manufacturing Firms in Daynile district Mogadishu to a large extent.

Table 1.5.3: Strategic Inventory Control and Quality

Quality	Mean	Std. Dev
Conformance to specifications	4.08	1.06
Reduced number of defects	3.92	1.18
Customer satisfaction	4.00	1.08
Customer service improvement	4.10	1.10
Improved brand name and reputation	4.02	1.11
Overall Score	**4.01**	**0.99**

Table 1.5.3: indicates that Strategic Inventory Control Techniques influence a firm quality by having products that conform to specifications (M= 4.08, SD= 1.06), having satisfied customers (M= 4.00, SD= 1.08) and improving customer service (M= 4.10, SD= 1.10) as indicated by the respective mean and SD. Improved brand name and reputation (M= 4.01, SD= 0.99) and reduced number of defects (M= 3.92, SD= 1-18) were also influenced to a large extent through the adoption of Strategic Inventory Control Techniques. The overall score (M= 4.01, SD= 0.99) implies that Strategic Inventory Control Techniques influences Quality to a large extent.

4.6 Relationship between Strategic Inventory Control and Operational Performance.

Table 1.6.1: Regression Coefficient Analysis

Pearson correlation	Operational performance	Strategic Inventory Control
Operational performance	1.00	.739
Strategic Inventory Control	.739	1.00

Table 1.6.1: indicates the results that there is high correlation between Strategic Inventory Control and Operational Performance with a value of 0.739 translating to 74%, so that independent variable (strategic inventory control) is associated with a regression coefficient describing the strength and the sign of that variable's relationship to the dependent variable (performance operation). This was important in measuring the extent to which changes in one or more variables jointly affected changes in another variable. The coefficients of Pearson correlation take values from +1 to -1 (0= no relationship and 1= a strong relationship between the variables).

Table 1.6.2: Regression Analysis of Strategic Inventory Control and Operational performance

Model	R	R square	Adjusted square	R Std. error of the estimate
1	.891	.693	.715	.194

Table 1.6.2: indicates that the coefficient of determination (R square) is 0.693 which translates to 69%. This therefore means that 69% of Operational Performance is attributed to Strategic Inventory Control. This is thus a significant fit as only 30.7% of the variation in Operational Performance is not accounted for. Therefore, further research should be conducted to assess the effects of strategic inventory control on operational performance.

Table 1.6.3: Analysis of Variance

Model	Sum of Squares	Df	Mean Square	R Std. error of the estimate	F	Sig.
1	Regression	7.392	4	1.233	3.898	.015[b]
Residual	0.837	26	0.064			
Total	8.229	30				

Table 1.6.3: The model significant value is 0.015 (F=3.898) and is less than the alpha value of 0.05 (5%). This implies that the regression model is statistically significant and that Strategic Inventory Control Techniques have a significant relationship with Operational performance.

Table 1.6.4: Significance of regression coefficients

Model	Unstandardized Coefficients		Standardized Coefficients	T	Sig.
	B	Std. Error	Beta		
1 (constant)	4.595	1.745		2.633	0.01
EOQ	0.509	0.143	0.344	3.559	0.01
JIT	0.201	0.068	0.563	2.956	0.00
ABC	0.397	0.08	0.439	4.963	0.02
WMS	0.215	0.035	0.376	6.143	0.02
Demand forecasting	0.314	0.166	0.436	1.890	0.03

Table 1.6.4: indicates that the established linear regression equation as per the SPSS generated table above, Y= a + β1X1+ β2X2+ β3X3+ β4X4 + β5X5 +e) that is Operational Performance (Y) = 4.595 + 0.509XEOQ + 0.201JIT + 0.397ABC + 0.215WMS +0.314 Demand Forecasting.

Strategic Inventory Control Techniques (Economic Order Quantity (t=03.559, P<0.05), Just-In-Time (t=2.956, P<0.05), ABC Analysis (t=4.963, P<0.05) Warehouse Management System (t=6.143, P<0.05) and Demand Forecasting (t=1.890, P<0.05) all have a positive and significant relationship with operational performance. The model shows that when all variables are held at constant, the value of Operational Performance would be 4.95.

From the above findings, Strategic Inventory Control Techniques were found to have statistically significant relationships with Operational Performance. The objective of determining the relationship between Strategic Inventory Control Techniques and Operational Performance has thus been achieved. The outcome of the second objective is supported by Stochastic Inventory Theory. This is because the theory will aid the manufacturing firms in ways that they can be able to avoid shortages of inventory and be able to meet customer demand on time. The theory will also help the firm in knowing how to keep records of stock when to replenish the stock and how much to replenish to meet the demands of their clients effectively and efficiently.

Discussions

This study was focused to investigate the influence of Strategic Inventory Control on operational Performance by manufacturing firms in Daynile district, Mogadishu, Somalia. The main objectives of the research were to find out the extent of Strategic Inventory Control adoption and to ascertain the correlation between Strategic Inventory Control and Operational Performance by manufacturing firms in Mogadishu, Somalia. The theories used in the study were Total Cost Analysis Theory and Stochastic Inventory Theory. The study methodology *used was descriptive and cross-sectional in design* and

sample size was 31 respondents. Descriptive statistics analyzed objective one of the studies while regression analysis analyzed objective two.

Manufacturing firms adopted EOQ to minimize storage and used a classical EOQ model to a large extent. Periodical replenishments of stock and maintaining minimum stock levels were also in place. It was also noted that JIT Inventory Control system was put in place to minimize waste and to ensure that enough materials are maintained just at the right place and time so as to reduce mishandling of materials. ABC Analysis was also in place so as to classify items based on their stock Value and to be able to reduce stock holding cost. The Technique was also used to set prices and manage stock levels. It was also determined that WMS played an important role in reviewing inventory levels on a timely basis and in detecting stock variance. Warehouse Management System was also in place to manage inbound and outbound logistics and at the same time aid in utilizing space of the warehouse. Keeping records for future references and taking into account forecast error played a vital role in forecasting of the demand. All these techniques played an important role in ensuring that operational performance has been enhanced in the manufacturing firms in Daynile district, Mogadishu, Somalia.

The study established that Economic Order Quantity, Just-In-Time, ABC Analysis, Warehouse Management System and Demand Forecasting were all adopted to a large extent by manufacturing firms in Daynile district, Mogadishu, Somalia. This is an indication that the first objective of the study has been achieved. It has also been established that there is a positive and substantial connection between Strategic Inventory Control and Operational Performance by Manufacturing Firms in Daynile district, Mogadishu, Somalia. Specifically, Efficiency, Speed and Quality have all been influenced by Strategic Inventory Control Techniques to a large extent and all have a positive and significant relationship with Strategic Inventory Control Techniques. This means that there existed a direct and noteworthy relationship amongst Independent Variable (Economic Order Quantity, Just-In-Time, ABC Analysis, Warehouse Management System and Demand Forecasting) and dependent Variable (Efficiency, Speed and Quality) This is an indication that the second objective of the study has also been fulfilled.

Conclusions

Based on the study findings, it is concluded that a significant results from this paper clearly indicates that Strategic Inventory Control Techniques significantly and positively influence Operational Performance in entities that have adopted the techniques. To be able to increase efficiency, improve the speed of serving customers and production and improve the quality of products and services, firms need to integrate Strategic Inventory Control Practices in their Strategy as it has been established that it impacts on Performance to a large extent. The improved performance has been reflected through reduced lead and cycle time, conformance to specifications, reduced overall cost, reduced delivery time, improved brand image and increased customer base due to customer satisfaction. Additionally, it can be resolved that entities that adopt EOQ, Just-In-Time, ABC Analysis, Warehouse Management System and Demand Forecasting are likely to achieve operational performance, gain a competitive edge and minimize the overall cost. The findings established that most of the Strategic Inventory Control Techniques were adopted to a large extent by the manufacturing firms in Daynile district, Mogadishu Somalia.

Recommendations

The findings established that most of the Strategic Inventory Control Techniques were adopted to a large extent by the manufacturing firms in Daynile district, Mogadishu Somalia. The study endorses that manufacturing firms should fully adopt Strategic Inventory Control Techniques as it has been established that it

has a positive correlation with Operational Performance.

The study recommends that all manufacturing firms should adopt EOQ, Just-In-Time, ABC Analysis, Warehouse Management System and Demand Forecasting in their processes for them to be Efficient, have speed and improve their Quality. This is due to the fact that it has been established that the adoption of Strategic Inventory Control Techniques influences Efficiency, Speed and Quality to a large extent. Therefore, firms who embrace Strategic Inventory Control Techniques are bound to enjoy these benefits. Improving inventory control strategies calls for a high degree of collaboration and visibility across all parties as well as utilizing sophisticated technologies. The study recommends that the management should constantly expose its staff to training in order to improve their skills on inventory management and enable the employees to understand the current inventory systems which when used will help the organization reduce on costs associated with holding inventory. There is also a need to create awareness on Inventory Management as a whole in the manufacturing industry and thus the study recommends that the government and stakeholders involved should create awareness on the relevance of Inventory Management to the manufacturing firms.

Suggestions for Further Research

Studies to be carried out in future need to explore why some practices are implemented to a large extent and why others are implemented to a medium extent. Other Strategic Inventory Control Techniques which are not covered in this study should also be investigated to establish the reason as to why they are not being adopted. Another study should be carried out to establish how Strategic Inventory Control Techniques influences organizational or supply chin Responsiveness, Procurement Performance or even how the Techniques gives an entity a competitive edge over their competitors. Future studies should also focus on Strategic Inventory Control Techniques in other sectors (retailing, third-party logistics, service industry) apart from Manufacturing Firms. Drivers and barriers of Strategic Inventory Control Techniques can also be another area for future studies.

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